

Group Seminar

Towards Feed-forward 3D Reconstruction

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Contents

- [1] Wang, Q., Zhang, Y., Holynski, A., Efros, A., & Kanazawa, A. (2025). Continuous 3D Perception Model with Persistent State. [\(CVPR 2025 Oral\)](#)
- [2] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, & Matt Feiszli (2025). Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass. [\(CVPR 2025\)](#)
- [3] Deng, J., Li, H., Xie, T., Ren, W., Zhang, Q., Tan, P., & Guo, X. (2025). SAIL-Recon: Large SfM by Augmenting Scene Regression with Localization.

Background - 3D Reconstruction

Deep Learning

Optimization-based

Incremental:
COLMAP

Global:
GLOMAP

Optimization-based

With Global Alignment:
DUST3R, MAST3R

Scene Coordinate
Reconstruction:
ACE0

Others:
FlowMap, VGGSfM

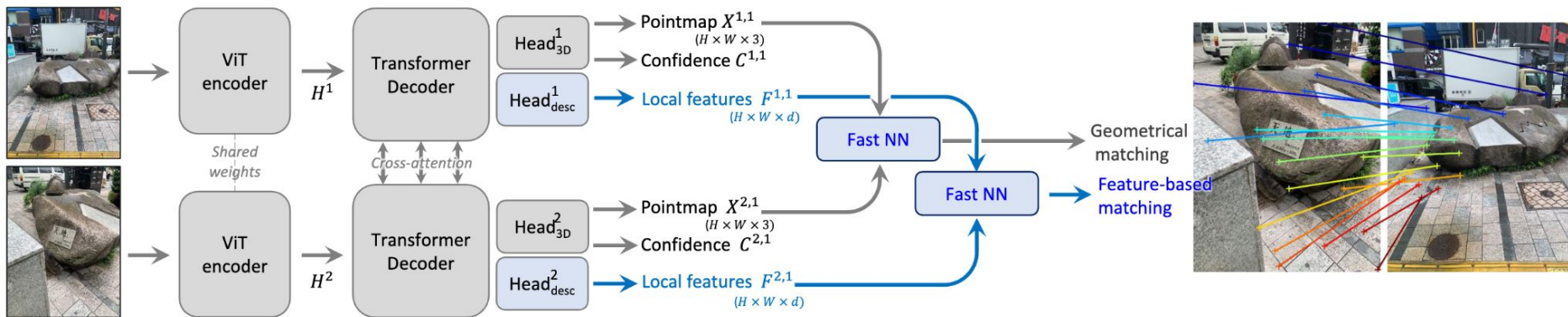
Feed-forward

Scene Regression,
VGGT-based:
SAIL-Recon

DUST3R-based:
Cut3R, Fast3R

Background - Basis

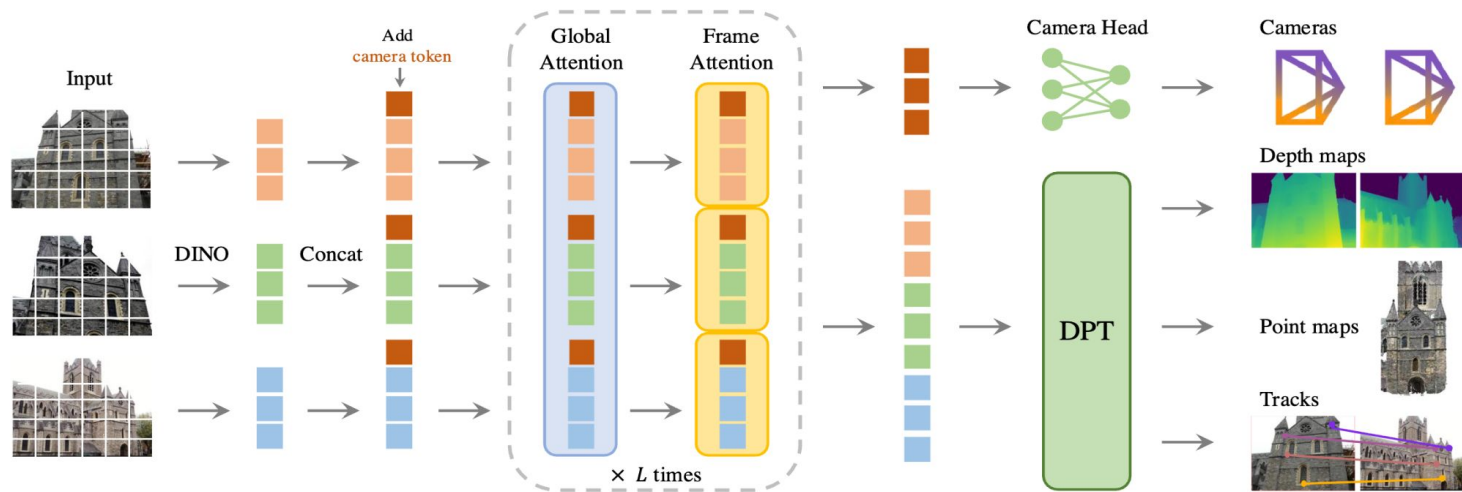
MASt3R (DUS3R)



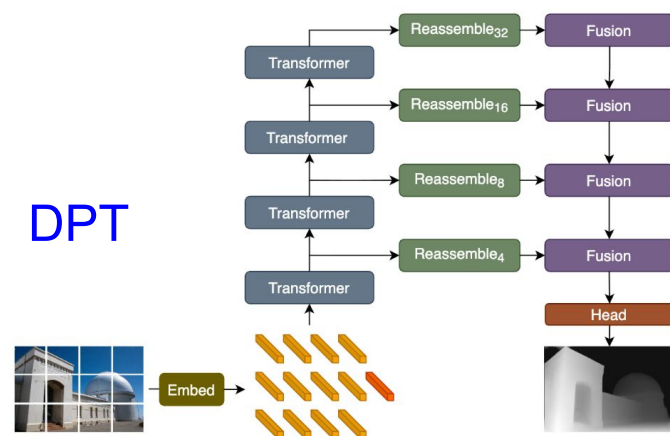
$$\chi^* = \arg \min_{\chi, P, \sigma} \sum_{e \in \mathcal{E}} \sum_{v \in e} \sum_{i=1}^{HW} C_i^{v,e} \|\chi_i^v - \sigma_e P_e X_i^{v,e}\|$$

Background - Basis

VG GT



DPT



Background - 3D Reconstruction

Deep Learning

Optimization-based

Incremental:
COLMAP

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GLOMAP

Optimization-based

With Global Alignment:
DUST3R, MAST3R

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Feed-forward

Scene Regression,
VGGT-based:
SAIL-Recon

DUST3R-based:
Cut3R, Fast3R

Background - Feed-forward SfM

Strengths

- Speed (cf. Bundle Adjustment, Features Matching)
- Learned priors from data
- Reduced error accumulation

Weaknesses

- Generalization
- Accuracy

Background - Current Problems & Perspectives

- Scalability (OOM)
 - Fast3R
- Large-scale 3D Reconstruction
 - SAIL-Recon
- Online 3D Reconstruction
 - Cut3R

Fast3R

- Scalability (OOM)

- **Fast3R**

Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

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Fast3R

- Observation

1. DUS3R leads to OOM with only 48 views on an A100 GPU
2. DUS3R computes $O(N^2)$ pairs of pointmaps and performs a global alignment optimization procedure
3. DUS3R suffers from the same **pair-wise bottleneck** as traditional SfM

Fast3R

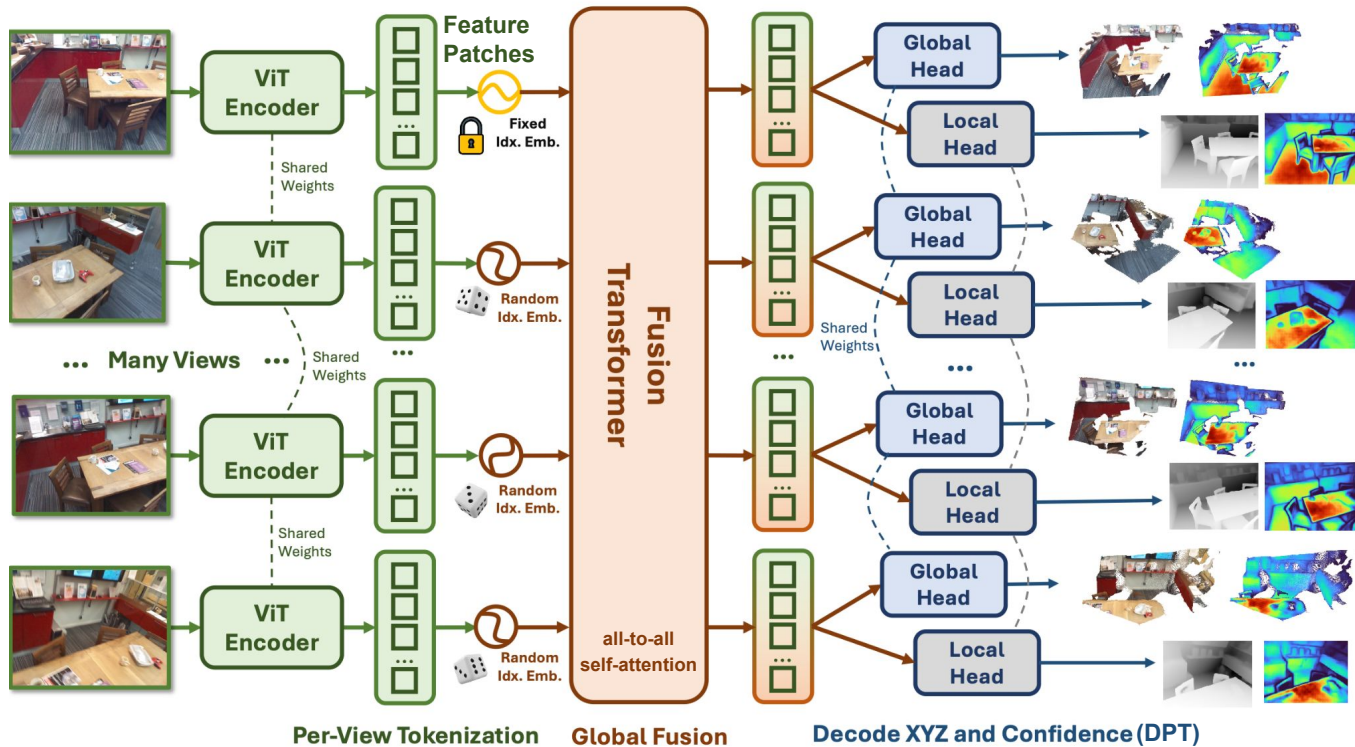
Inputs: **Unordered**, unposed, RGB images

$$I \rightarrow (\mathbf{X}_L, \Sigma_L, \mathbf{X}_G, \Sigma_G)$$

Global Pointmap $\mathbf{X}_G$:
In the coordinate
frame of the 1st
camera

Local Pointmap $\mathbf{X}_L$:
In the coordinate
frame of the viewing
camera

Align $\mathbf{X}_L$ to $\mathbf{X}_G$ using ICP
=> aligned local pointmaps
=> 3D reconstruction



Fast3R - Technical Highlights

- Training Objective: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\mathbf{X}_G} + \mathcal{L}_{\mathbf{X}_L}$; **Confidence-weighted** versions of the normalized 3D pointwise regression loss

$$\ell_{\text{regr}}(\hat{\mathbf{X}}, \mathbf{X}) = \left\| \frac{1}{\hat{z}} \hat{\mathbf{X}} - \frac{1}{z} \mathbf{X} \right\|_2, \quad z = \frac{1}{|\mathbf{X}|} \sum_{x \in \mathbf{X}} \|x\|_2$$

Dataset: CO3D, ScanNet++, ARKitScenes, Habitat, **BlendedMVS**, **MegaDepth**
(less than DUST3R)

- Fast3R can handle more views at inference than were used to train a model

Naive way: Spherical Harmonic

- Training: $\text{SH}(\{1, 2, \dots, N\})$
- Testing: $\text{SH}(\{1, 2, \dots, N_{\text{test}}\})$

Adoption: Position Interpolation

- Draw $N \subset \{1, 2, \dots, N'\}$
indexes
uniformly at random ($N' \gg N$)
Masking!

Fast3R - Experiments

Inference Efficiency

# Views	Fast3R		DUST3R	
	Time (s)	Peak GPU Mem (GiB)	Time (s)	Peak GPU Mem (GiB)
2	0.065	3.84	0.092	3.52
8	0.122	6.33	8.386	24.59
32	0.509	13.25	129.0	67.61
48	0.84	20.8	OOM	OOM
320	15.938	41.90	OOM	OOM
800	89.569	55.97	OOM	OOM
1000	137.62	63.01	OOM	OOM
1500	308.85	78.59	OOM	OOM

3D Reconstruction

Method	FPS	7 scenes [47]		NRGBD [3]	
		Acc↓	Comp↓	Acc↓	Comp↓
F-Recon [65]	<0.1	7.62	2.31	20.59	6.31
DUST3R [†] [61]	0.78	1.23	0.91	2.51	1.03
Spann3R [57]	65.4	1.48	0.85	3.15	1.10
Fast3R (Ours)	251.1	1.58	0.93	3.40	1.01
		Skip = 20		Skip = 40	
Method	Views	Acc↓	DTU	Comp↓	
		Med.		Med.	
DUST3R [61]	all/5	1.159		0.914	
DUST3R [†] [61]	all/5	1.297		1.002	
Spann3R [57]	all/5	2.268		1.295	
Fast3R (Ours)	all/5	1.706		0.857	

Fast3R - Experiments

Pose Estimation

Methods	CO3Dv2 [39]					RealEstate10K [75]	FPS
	RRA@15↑	RRA@5↑	RTA@15↑	RTA@5↑	mAA(30)↑	mAA(30)↑	
Colmap+SG [9, 41]	36.1	24.4	27.3	17.2	25.3	45.2	0.056
PixSfM [28]	33.7	26.1	32.9	17.6	30.1	49.4	-
RelPose [71]	57.1	-	-	-	-	-	0.02
PosReg [59]	53.2	-	49.1	-	45.0	-	0.015
PoseDiff [59]	80.5	59.5	79.8	61.7	66.5	48.0	0.015
RelPose++ [27]	(85.5)	-	-	-	-	-	0.02
DUST3R [60]	96.2	-	86.8	-	76.7	67.7	0.78
MASt3R [23]	94.6	93.2	91.9	86.2	81.8	76.4	0.23
Fast3R-no-outdoor (Ours)	99.7	97.4	87.1	76.1	82.5	-	251.1
Fast3R (Ours)	96.2	90.2	81.6	68.2	75.0	72.7	251.1

SAIL-Recon

- Large-scale 3D Reconstruction

- **SAIL-Recon**

Based on VGGT

SAIL-Recon: Large SfM by Augmenting Scene Regression with Localization

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Ping Tan^{1†} Xiaoyang Guo^{2†}

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SAIL-Recon

- Observation
 1. VGGT struggle to handle a large number of input images
 2. Current large-scale scene regression (SCR) methods suffer from pose drifting and are highly dependent on subsequent global alignment (e.g. online, divide and conquer methods)
 3. Existing SCR methods ignore visual localization, a common approach to scale up an SLAM system

SAIL-Recon

Inputs: **Unordered**, unposed, RGB images

$$(\mathcal{R}, \{T_i, K_i, D_i, S_i\}_{i=1}^M) = \mathcal{T}_\theta(\{\mathcal{I}_i\}_{i=1}^M)$$

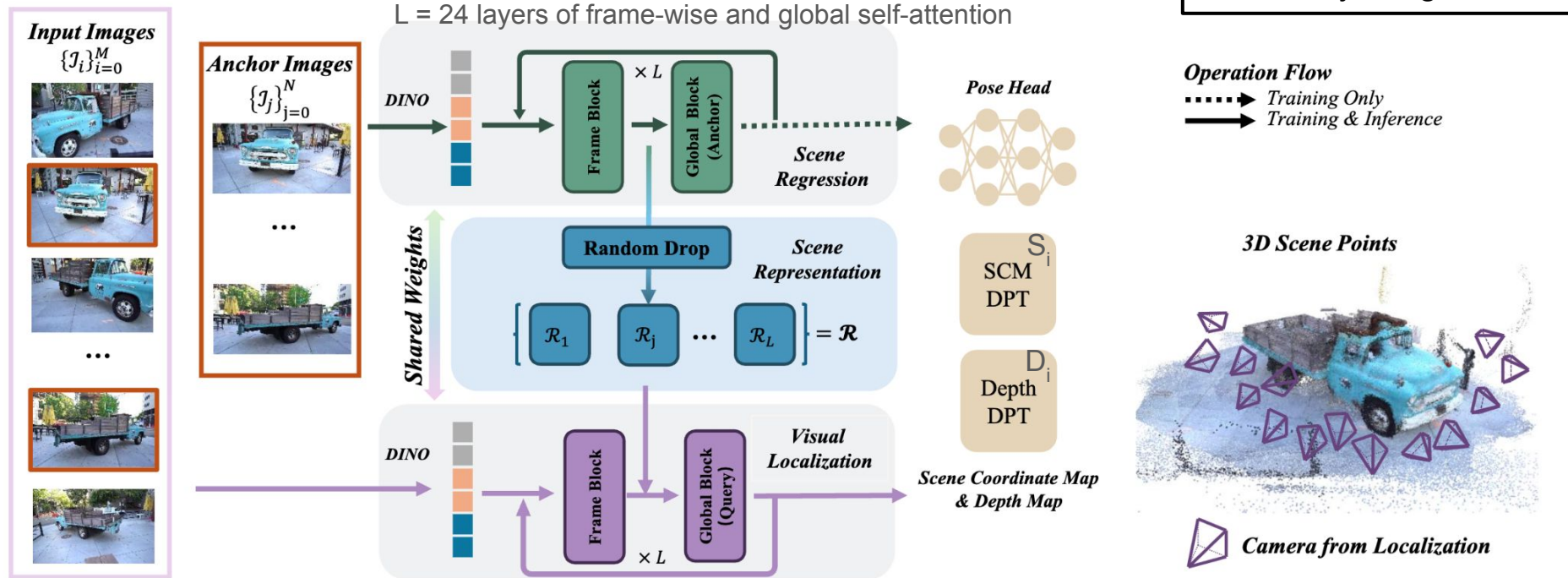
$$\{T^q, K^q, D^q, S^q\} = \mathcal{T}_\theta(\mathcal{I}^q, \mathcal{R})$$

$\mathcal{T}_\theta$: Transformer-based network

T_i : Extrinsic matrices

K_i : Intrinsic matrices

$\mathcal{I}^q$: Query image



SAIL-Recon - Technical Highlights

- Neural scene representation serves as an **implicit** neural map of the scene, without relying on explicit 3D points or meshes
- Most localization methods require per-scene training of the network
=> SAIL-Recon circumvents this by deriving a latent scene representation through scene regression
- Uniformly sample $N \in [50, 100]$ anchor frames $\{\mathcal{I}_i\}_{i=1}^N$ from the full set $\{\mathcal{I}_i\}_{i=1}^M$

SAIL-Recon - Technical Highlights

The network $\mathcal{T}_\theta$ takes a set of 2D images as input and progressively enhances them to compute 3D structures

=> Extract **intermediate feature tokens from each attention layer** of $\mathcal{T}_\theta$ to form the scene representation $\mathcal{R}$

(Captures the gradual evolution from 2D appearance features to 3D coordinate descriptors)

$$\mathcal{R} = [\{\Theta(t_j'^{\mathcal{I}})\}_{j=1}^L]$$

Θ : Downsampling operation
 $t_j'^{\mathcal{I}}$: Intermediate feature tokens

Randomly select a ratio $r \in [0.2, 1.0]$

=> Sample a subset of tokens in each anchor frame to control the total size of $\Theta(t_j'^{\mathcal{I}})$

SAIL-Recon - Technical Highlights

Method	Co3Dv2 [↑]			
	mAA@30	mAA@5	RRA@15	RTA@15
SAIL-Recon	87.3	57.6	98.3	93.6
300 per image → Fixed token	86.5	53.6	98.2	93.6
Avg. Pooling	86.5	53.5	98.2	93.5

Table 7. **Ablation on Training Strategy.** We investigate the different training strategies on pose estimation accuracy.

$$\mathcal{R} = [\{\Theta(t_j'^{\mathcal{I}})\}_{j=1}^L]$$

Θ : Downsampling operation
 $t_j'^{\mathcal{I}}$: Intermidate feature tokens

Randomly select a ratio $r \in [0.2, 1.0]$

=> Sample a subset of tokens in each anchor frame to control the total size of $\Theta(t_j'^{\mathcal{I}})$

SAIL-Recon - Technical Highlights

- Training Objective: $\mathcal{L} = \mathcal{L}_{\text{camera}} + \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scm}}$
(**Fine-tuning** VGGT checkpoints)

$$\mathcal{L}_{\text{camera}} = \sum_{i=1}^N \|\hat{g}_i - g_i\|_1$$

$$\mathcal{L}_{\text{depth}} = \sum_{i=1}^N \|C_i^D \odot (\hat{D}_i - D_i)\| + \|(\nabla \hat{D}_i - \nabla D_i)\| - \alpha \log C_i^D$$


Gradient-based smooth

$$\mathcal{L}_{\text{scm}} = \sum_{i=1}^N \|C_i^S \odot (\hat{S}_i - S_i)\| + \|(\nabla \hat{S}_i - \nabla S_i)\| - \alpha \log C_i^S$$

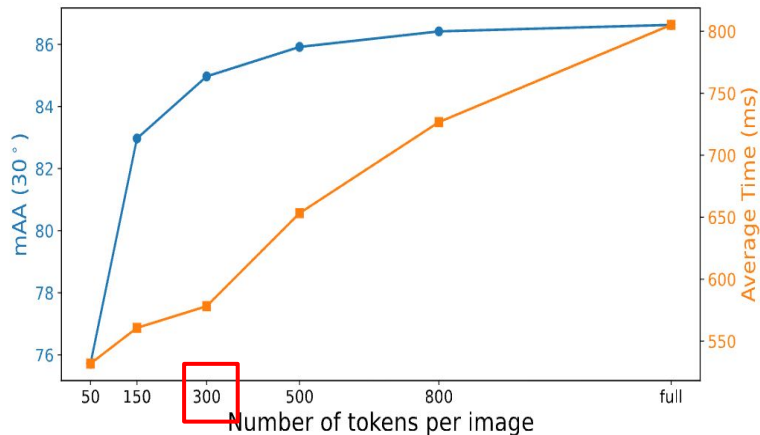
C_i^D : Uncertainty Map
 $g_i = [\mathbf{q}, \mathbf{t}, \mathbf{f}]$:
Camera Parameters
 $\mathbf{q}$: Camera orientation
 $\mathbf{t}$: Translation vector
 $\mathbf{f}$: Field of View

With **normalization** w.r.t. a randomly selected image in anchor frames

SAIL-Recon - Experiments

Pose Accuracy - Tanks & Temples.

Method	Align.	RRA@5 $\uparrow$	RTA@5 $\uparrow$	ATE $\downarrow$	Reg. $\uparrow$	Time [s] $\downarrow$
COLMAP [60]	OPT	GT	GT	GT	GT	-
GLOMAP [48]	OPT	75.8	76.7	0.010	100.0	1977
ACE0 [9]	OPT	56.9	57.9	0.015	100.0	5499
DF-SfM [26]	OPT	69.6	69.3	0.014	76.2	-
FlowMap [63]	OPT	31.7	35.7	0.017	66.7	-
VGGsFm [84]	OPT	-	-	-	0.0	2134
MASt3R-SfM [19]	OPT	49.2	54.0	0.011	100.0	2723
DROID-SLAM [75]	OPT	31.3	40.3	0.021	100.0	240
SAIL-Recon-OPT	OPT	71.5	77.7	0.008	100.0	233
Cut3R [†] [86]	FFD	18.8	25.8	0.017	100.0	42
Spann3R [†] [82]	FFD	22.1	30.7	0.016	100.0	116
SLAM3R [†] [42]	FFD	20.3	24.7	0.015	100.0	70
Light3R-SfM [21]	FFD	52.0	52.8	0.011	100.0	63
VGGT-SLAM* ^{Δ} [43]	FFD	57.3	67.9	0.008	100.0	238
SAIL-Recon	FFD	70.4	74.7	0.008	100.0	81



The evaluation of camera poses is sometimes **unreliable**, as the **pseudo ground-truth** from COLMAP is only an estimation!

SAIL-Recon - Experiments

Pose accuracy via **view synthesis on** Mip-NeRF 360

PSNR

	Pseudo GT COLMAP	DROID- SLAM [†] [75]	BARF [37]	Nope- NeRF [5]	ACE0 [9]	Ours
Bicycle	21.5	10.9	11.9	12.2	18.7	20.50
Bonsai	27.6	10.9	12.5	14.8	25.8	26.76
Counter	25.5	12.9	11.9	11.6	24.5	25.51
Garden	26.3	16.7	13.3	13.8	25.0	24.92
Kitchen	27.4	13.9	13.3	14.4	26.1	27.43
Room	28.0	11.3	11.9	14.3	19.8	27.46
Stump	16.8	13.9	15.0	13.9	20.5	20.83
Average	24.7	12.9	12.8	13.5	22.9	24.77
Average Time	1h	2min	4h	≥24h	8h	5min

SAIL-Recon - Ablation

Method	Global [†] Align.	Co3Dv2 [‡]		
		RRA@15	RTA@15	mAA@30
Colmap [59]	OPT	31.6	27.3	25.3
Glomap [48]	OPT	45.9	40.3	37.3
PixSfM [38]	OPT	33.7	32.9	30.1
VGGSfM [84]	OPT	92.1	88.3	74.0
DUS3R-GA [87]	OPT	96.2	86.8	76.7
MASt3R-SfM [19]	OPT	96.0	93.1	88.0
PoseDiff [83]	FFD	80.5	79.8	66.5
RelPose++ [36]	FFD	82.3	77.2	65.1
Spann3R [82]	FFD	89.5	83.2	70.3
MASt3R * [34]	FFD	94.5	80.9	68.7
Light3R-SfM [21]	FFD	94.7	85.8	72.8
VGGT [85]	FFD	98.4	94.8	88.2
SAIL-Recon $N = 10$	FFD	98.3	94.0	88.1
SAIL-Recon $N = 8$	FFD	98.2	93.6	87.3
SAIL-Recon $N = 5$	FFD	97.7	92.2	85.0
SAIL-Recon $N = 2$	FFD	96.4	89.7	78.5

Ablation on the **number of anchor views** for pose estimation performance on **CO3Dv2**

of input images: 10

Background - Problems & Perspectives

- Online 3D Reconstruction

- **Cut3R**

Continuous 3D Perception Model with Persistent State

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¹University of California, Berkeley

²Google DeepMind

Cut3R

- Observation

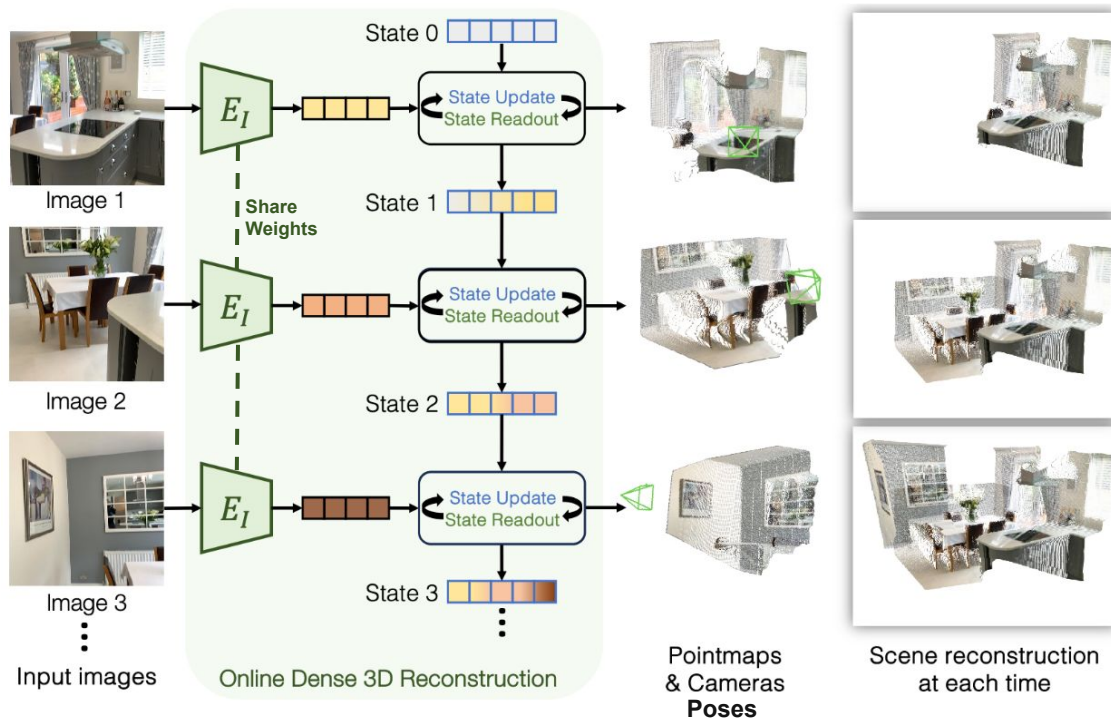
1. Many 3D reconstruction pipelines operate as tabula rasa, starting **from scratch** for each new scene (e.g. SfM, SLAM, ...)

=> poses challenges in handling scenarios with sparse observations

2. Current continuous reconstruction methods have limitation such as being object-centric, or limited to **static** scenes

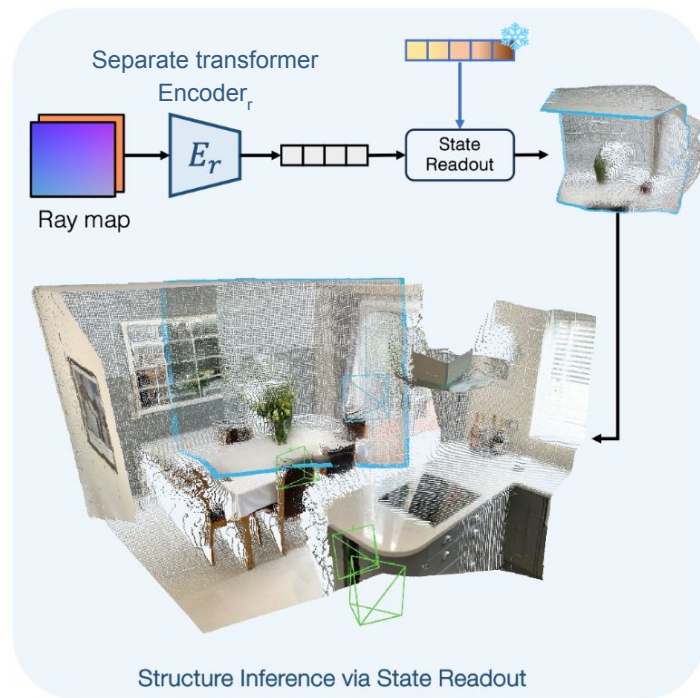
Cut3R

Inputs: Stream of images without any camera information



Ray map R :

A 6-channel image encoding the origin and direction of rays at each pixel



Cut3R - Technical Highlights

- The bidirectional interaction (update and readout) is implemented using two **interconnected** transformer decoders

$$[z'_t, F'_t], s_t = \text{Decoders}([z, F_t], s_{t-1})$$

$$\begin{aligned} \hat{X}_t^{\text{self}}, C_t^{\text{self}} &= \text{Head}_{\text{self}}(F'_t) \\ \hat{X}_t^{\text{world}}, C_t^{\text{world}} &= \text{Head}_{\text{world}}(F'_t, z'_t) \\ \hat{P}_t &= \text{Head}_{\text{pose}}(z'_t), \quad \text{MLP} \end{aligned} \quad \left. \vphantom{\begin{aligned} \hat{X}_t^{\text{self}}, C_t^{\text{self}} &= \text{Head}_{\text{self}}(F'_t) \\ \hat{X}_t^{\text{world}}, C_t^{\text{world}} &= \text{Head}_{\text{world}}(F'_t, z'_t) \end{aligned}} \right\} \text{DPT}$$

F_t : Image tokens
 z : Learnable pose token
 X_t : Pointmap
 C_t : Confidence map
 P_t : (6-DoF) Relative transformation

- Redundancy** simplifies training (i.e. predicting $\hat{X}_t^{\text{self}}$, $\hat{X}_t^{\text{world}}$, and $\hat{P}_t$)
- All pointmaps and poses are in **metric scale** (i.e., meters)

Cut3R - Technical Highlights

- Training Objective

$$\mathcal{L}_{\text{conf}} = \sum_{(\hat{\mathbf{x}}, c) \in (\hat{\mathcal{X}}, \mathcal{C})} \left(c \cdot \left\| \frac{\hat{\mathbf{x}}}{\hat{s}} - \frac{\mathbf{x}}{s} \right\|_2 - \alpha \log c \right)$$

Confidence-aware regression loss to the pointmaps. (When GT pointmaps are **metric**: set $\hat{s} := s$)

$$\mathcal{L}_{\text{pose}} = \sum_{t=1}^N \left(\left\| \hat{\mathbf{q}}_t - \mathbf{q}_t \right\|_2 + \left\| \frac{\hat{\boldsymbol{\tau}}_t}{\hat{s}} - \frac{\boldsymbol{\tau}_t}{s} \right\|_2 \right)$$

Pose loss. We parameterize the pose $\hat{P}_t$ as quaternion $\hat{\mathbf{q}}_t$ and translation $\hat{\boldsymbol{\tau}}_t$, and minimize the L2 norm between the prediction and groundtruth:

$$\mathcal{L}_{rgb} = \left\| \hat{\mathbf{I}}_r - \mathbf{I}_r \right\|_2^2$$

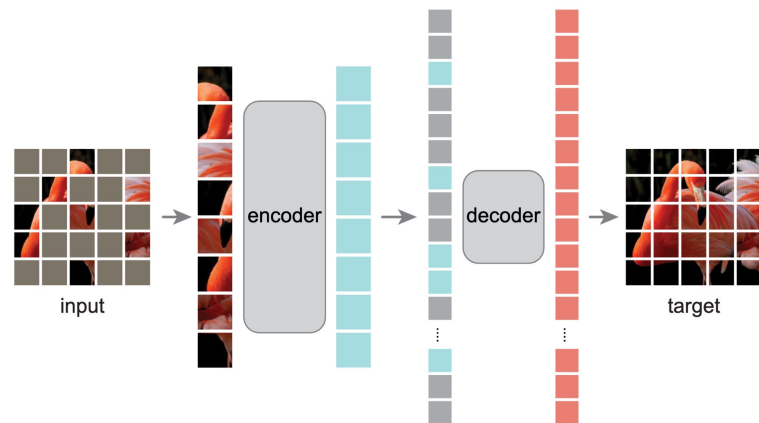
MSE loss. When the input is raymap, we also apply MSE loss to enforce the predicted pixel colors $\hat{\mathbf{I}}_r$ to match the groundtruth

Cut3R - Analogy

Cut3R

- Completion occurs at the **image** level
- Using global context of the **3D scene** captured in the state.

Masked Autoencoders



- Completion occurs at the **patch** level
- Using global context of the entire **image**

Cut3R - Experiments

Video Depth Estimation

Alignment	Method	Optim. Onl.	Sintel		BONN		KITTI		FPS
			Abs Rel ↓	$\delta < 1.25$ ↑	Abs Rel ↓	$\delta < 1.25$ ↑	Abs Rel ↓	$\delta < 1.25$ ↑	
Per-sequence scale	DUS _t 3R-GA [107]	✓	0.656	45.2	0.155	83.3	<u>0.144</u>	<u>81.3</u>	0.76
	MASt ₃ R-GA [51]	✓	0.641	43.9	0.252	70.1	0.183	74.5	0.31
	MonST ₃ R-GA [125]	✓	0.378	55.8	0.067	96.3	0.168	74.4	0.35
	Spann ₃ R [101]	✓	0.622	42.6	0.144	81.3	0.198	73.7	13.55
	Ours	✓	<u>0.421</u>	<u>47.9</u>	<u>0.078</u>	<u>93.7</u>	0.118	88.1	16.58
Metric scale	MASt ₃ R-GA [51]	✓	1.022	14.3	0.272	70.6	0.467	15.2	0.31
	Ours	✓	1.029	23.8	0.103	88.5	0.122	85.5	16.58

1. Absolute relative error (Abs Rel)
2. $\delta < 1.25$: Percentage of predicted depths within a 1.25-factor of true depth

Cut3R - Experiments

Camera Pose Estimation

Method	Optim.	Onl.	Sintel			TUM-dynamics			ScanNet		
			ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓
Particle-SfM [129]	✓		<u>0.129</u>	0.031	0.535	-	-	-	0.136	0.023	0.836
Robust-CVD [47]	✓		0.360	0.154	3.443	0.153	0.026	3.528	0.227	0.064	7.374
CasualSAM [128]	✓		0.141	0.035	<u>0.615</u>	<u>0.071</u>	0.010	1.712	0.158	0.034	1.618
DUST3R-GA [107]	✓		0.417	0.250	5.796	0.083	0.017	3.567	0.081	0.028	0.784
MASt3R-GA [51]	✓		0.185	0.060	1.496	0.038	<u>0.012</u>	0.448	<u>0.078</u>	<u>0.020</u>	0.475
MonST3R-GA [125]	✓		0.111	<u>0.044</u>	0.869	0.098	0.019	<u>0.935</u>	0.077	0.018	<u>0.529</u>
DUST3R [107]		✓	<u>0.290</u>	0.132	7.869	0.140	0.106	3.286	0.246	0.108	8.210
Spann3R [101]		✓	0.329	<u>0.110</u>	<u>4.471</u>	<u>0.056</u>	<u>0.021</u>	<u>0.591</u>	0.096	<u>0.023</u>	<u>0.661</u>
Ours		✓	0.213	0.066	0.621	0.046	0.015	0.473	<u>0.099</u>	0.022	0.600

		7 scenes [83]						NRGBD [4]						3D Reconstruction	
		Acc↓		Comp↓		NC↑		Acc↓		Comp↓		NC↑			
Method	Optim. Onl.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	FPS	
DUST3R-GA [107]	✓	<u>0.146</u>	<u>0.077</u>	0.181	<u>0.067</u>	0.736	0.839	0.144	0.019	0.154	0.018	0.870	0.982	0.68	
MASt3R-GA [51]	✓	0.185	0.081	<u>0.180</u>	0.069	0.701	0.792	0.085	0.033	0.063	0.028	0.794	0.928	0.34	
MonST3R-GA [51]	✓	0.248	0.185	0.266	0.167	0.672	0.759	0.272	0.114	0.287	0.110	0.758	0.843	0.39	
Spann3R [101]	✓	0.298	0.226	0.205	0.112	0.650	0.730	0.416	0.323	0.417	0.285	0.684	0.789	12.97	
Ours	✓	0.126	0.047	0.154	0.031	<u>0.727</u>	<u>0.834</u>	<u>0.099</u>	<u>0.031</u>	<u>0.076</u>	<u>0.026</u>	<u>0.837</u>	<u>0.971</u>	17.00	

Demo: Sail-Recon

